Samuel Kirschner

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Education

Massachusetts Institute of Technology — Cambridge, MA
Candidate for MS Mechanical Engineering — Sept. 2025 - Present

Georgia Institute of Technology — Metz, France | Atlanta, GA
BS Mechanical Engineering, GPA 4.0 — Jan. 2022 - May 2025

Experience

Therapeutic Technology Design & Development Lab — Cambridge, MA
Fabrication-Integrated Design Lab
Graduate Researcher — Sept. 2025 - Present

- Developing a framework for quantitative comparison of pressure generation technologies in soft robotics and medical devices, identifying hybrid architectures as a high-potential area for performance gains.
- Designing and prototyping a hybrid pressure generator coupling electrolytic gas generation with H_2O_2 monopropellant decomposition, exceeding the pressure generation limits of existing monopropellant systems for untethered soft robotics.

NASA Johnson Space Center — Houston, Tx
Engineering Intern — June. 2024 - Aug. 2024

- Advanced systems models for in-situ resource utilization processes and key components critical to establishing and sustaining permanent settlements on the Moon and Mars.
- Collaborated with the Energy Systems Division to define mission requirements for a lunar water mining and processing system, integrating high-level strategic directives with technical insights.
- Conducted a trade study of MBSE tools and presented findings to division leadership, streamlining software usage and influencing the strategic direction of future projects.

Physiology of Wearable Robotics Lab | Geometry of Materials Lab — Atlanta, Ga
Undergraduate Researcher — Jan. 2022 - May 2025

- Collaborated in a multidisciplinary team developing pseudo-active programmable wearable devices to enhance biomechanics and co-authored the first paper studying their potential.
- Performed experimental characterization of human biomechanics using motion capture, EMG, and torque-angle testing to develop a framework for optimization-based wearable device design.
- Studied the unique mechanical behavior of elephant skin using digital image correlation to provide deeper insight into soft-tissue mechanics for application in biomimetic materials and robotics.

NASA Marshall Space Flight Center — Huntsville, Al
Engineering Intern — June. 2022 - Aug. 2022

- Developed a model of lunar surface operations in MagicDraw, which was adopted by the Architecture and Baseline Definitions team for Artemis 3, the first crewed return to the moon.
- Integrated timing simulation capabilities into the Lunar Exploration Architecture Framework using SysML that was used to predict total event duration and probability of mission success.

Peer-Reviewed Publications

1. Singal, K.*, **Kirschner, S.P.***, Schulz, A.K., Sanders, E.D., Sawicki, G., Herrin, K.R., Matsumoto, E.A. "Augmenting knee biomechanics through programmable knitted ExoSkin orthoses," *arXiv:2509.22971.*

Skills

Design: CAD (SolidWorks, Fusion 360), DFM, Tolerance Analysis

Analysis and Modeling: MATLAB, Mathematica, Model-Based Systems Engineering (SysML)

Manufacturing: Additive Manufacturing, Injection Molding, Milling, Lathe, Waterjet, Statistical Process Control

Experimental Characterization: Precision Metrology, Kinematic Analysis, Optical Strain Measurement